replogle-analysis-highly-expressed

2026-03-19

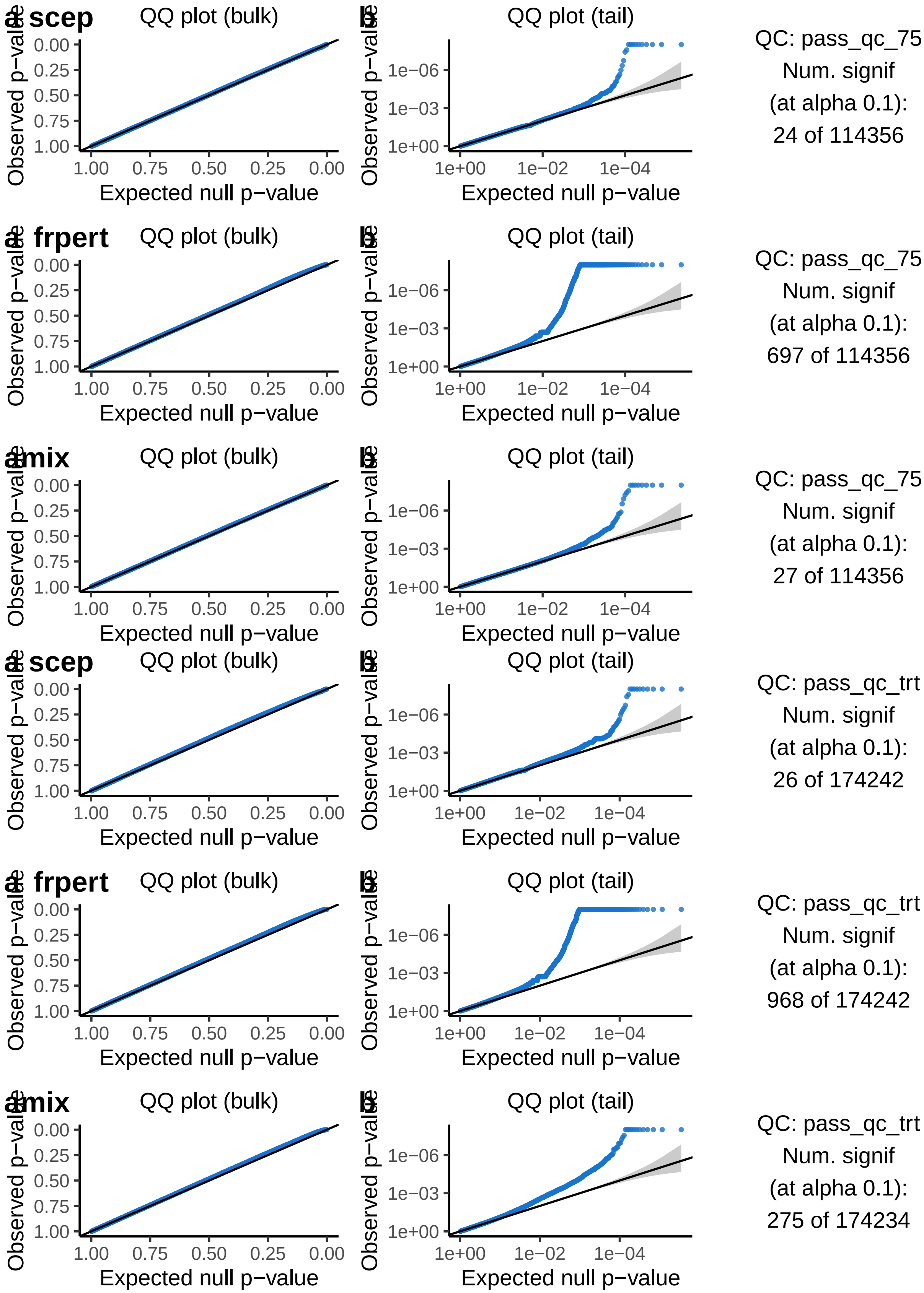
This notebook makes figures to explore the following:

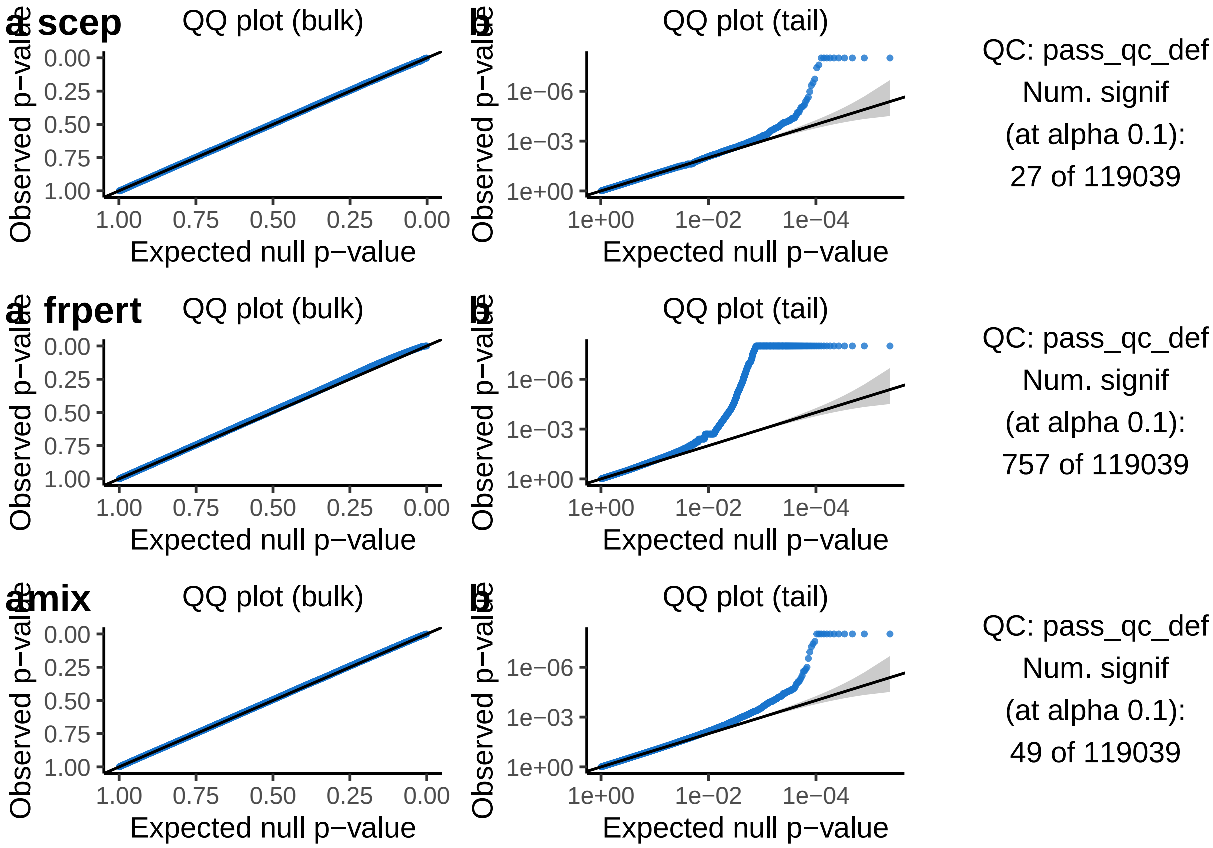
analysis of positive and negative control on replogle, only using highly expressed genes.

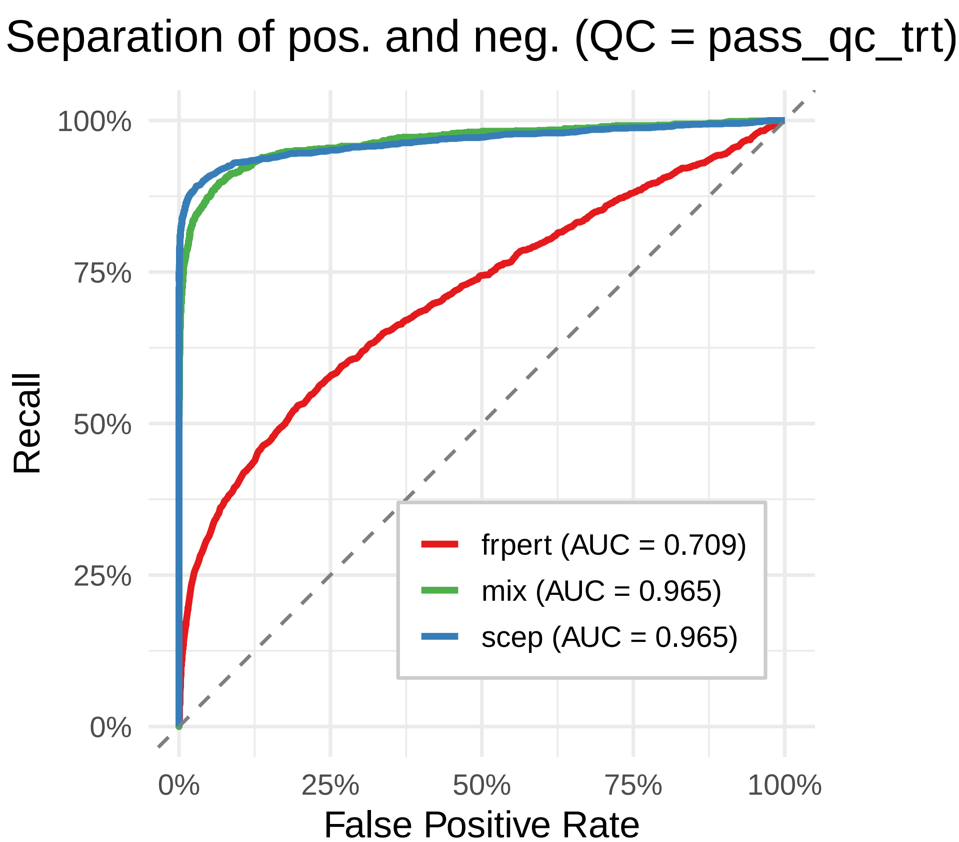
```
## Dataset: replogle-rd7_neg_control_4000genes : #genes = ; #cells = 13548
## #guides = 113 ; #targets = 113 ; avg # guides / target: 1
## # cells per guide: median = 111 ; min = 29 ; max = 617
## # cells per target: avg = 111 ; min = 29 ; max = 617
## # guides per cell: moi (avg) = 1 ; min = 1 ; max = 1
## # discovery pairs: 293348
##
##  ****************************************************************
## Dataset: replogle-rd7_pos_control_large : #genes = 1651 ; #cells = 210084
## #guides = 1854 ; #targets = 1652 ; avg # guides / target: 1.12
## # cells per guide: median = 86 ; min = 8 ; max = 1766
## # cells per target: avg = 87 ; min = 20 ; max = 10058
## # guides per cell: moi (avg) = 1 ; min = 1 ; max = 1
```

```
## In total: 1651 pos. ctrl pairs and 293348 neg. ctrl pairs.
```

```
##       pass_qc_75 pass_qc_trt pass_qc_def
## FALSE     179170      119344      175240
## TRUE      115829      175655      119759
```


a scep
QQ plot (bulk)
Observed p-value
Expected null p−value
b
QQ plot (tail)
QC: pass_qc_75
Num. signif
(at alpha 0.1):
24 of 114356
a frpert
QC: pass_qc_75
Num. signif
(at alpha 0.1):
697 of 114356
a mix
QC: pass_qc_75
Num. signif
(at alpha 0.1):
27 of 114356
a scep
QC: pass_qc_trt
Num. signif
(at alpha 0.1):
26 of 174242
a frpert
QC: pass_qc_trt
Num. signif
(at alpha 0.1):
968 of 174242
a mix
QC: pass_qc_trt
Num. signif
(at alpha 0.1):
275 of 174234


a scep
QQ plot (bulk)
Observed p−value
Expected null p−value
b
QQ plot (tail)
QC: pass_qc_def
Num. signif
(at alpha 0.1):
27 of 119039
a frpert
QQ plot (bulk)
b
QQ plot (tail)
QC: pass_qc_def
Num. signif
(at alpha 0.1):
757 of 119039
a mix
QQ plot (bulk)
b
QQ plot (tail)
QC: pass_qc_def
Num. signif
(at alpha 0.1):
49 of 119039


Separation of pos. and neg. (QC = pass_qc_trt)
Recall
False Positive Rate
frpert (AUC = 0.709)
mix (AUC = 0.965)
scep (AUC = 0.965)

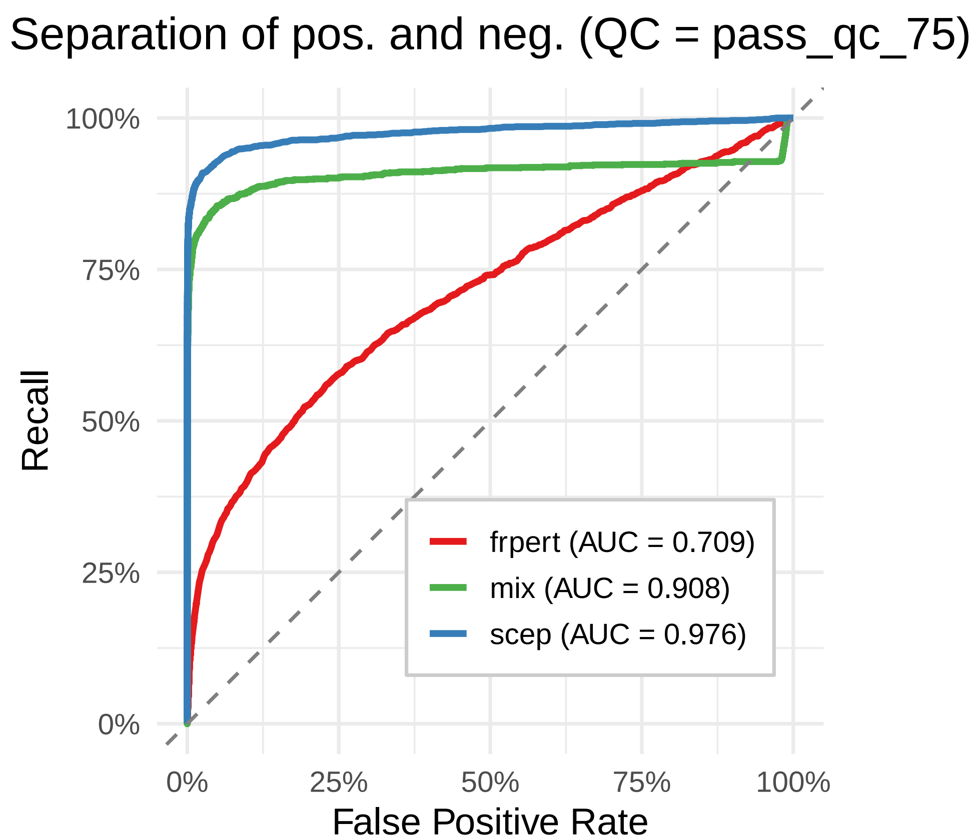
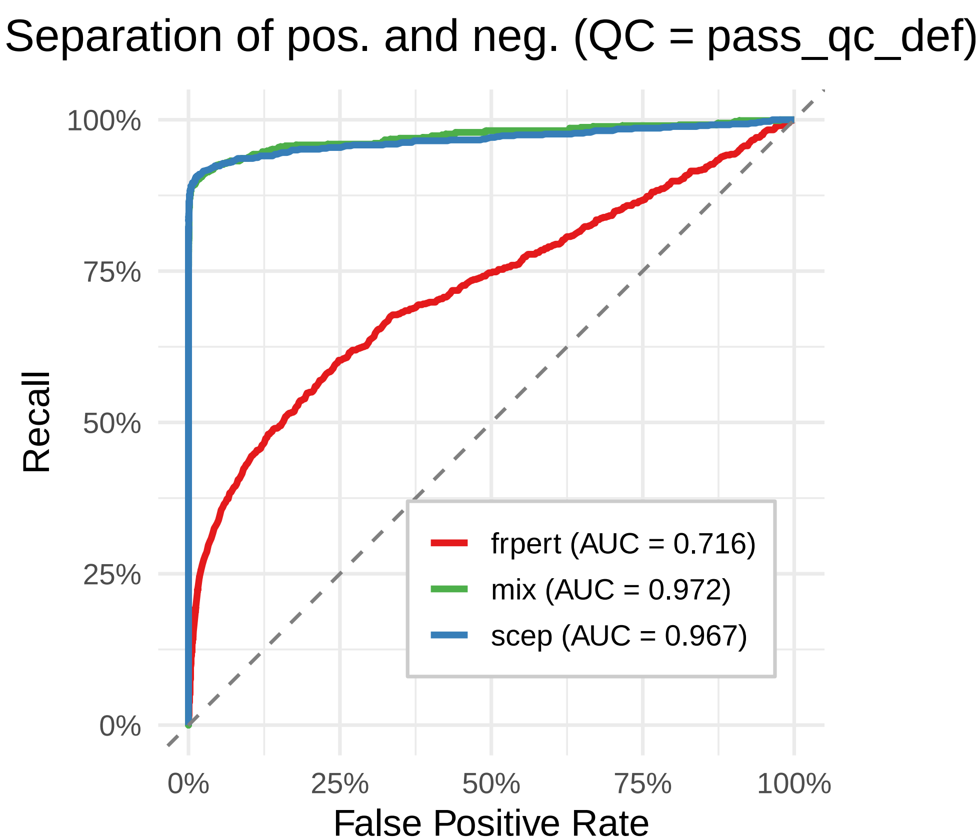
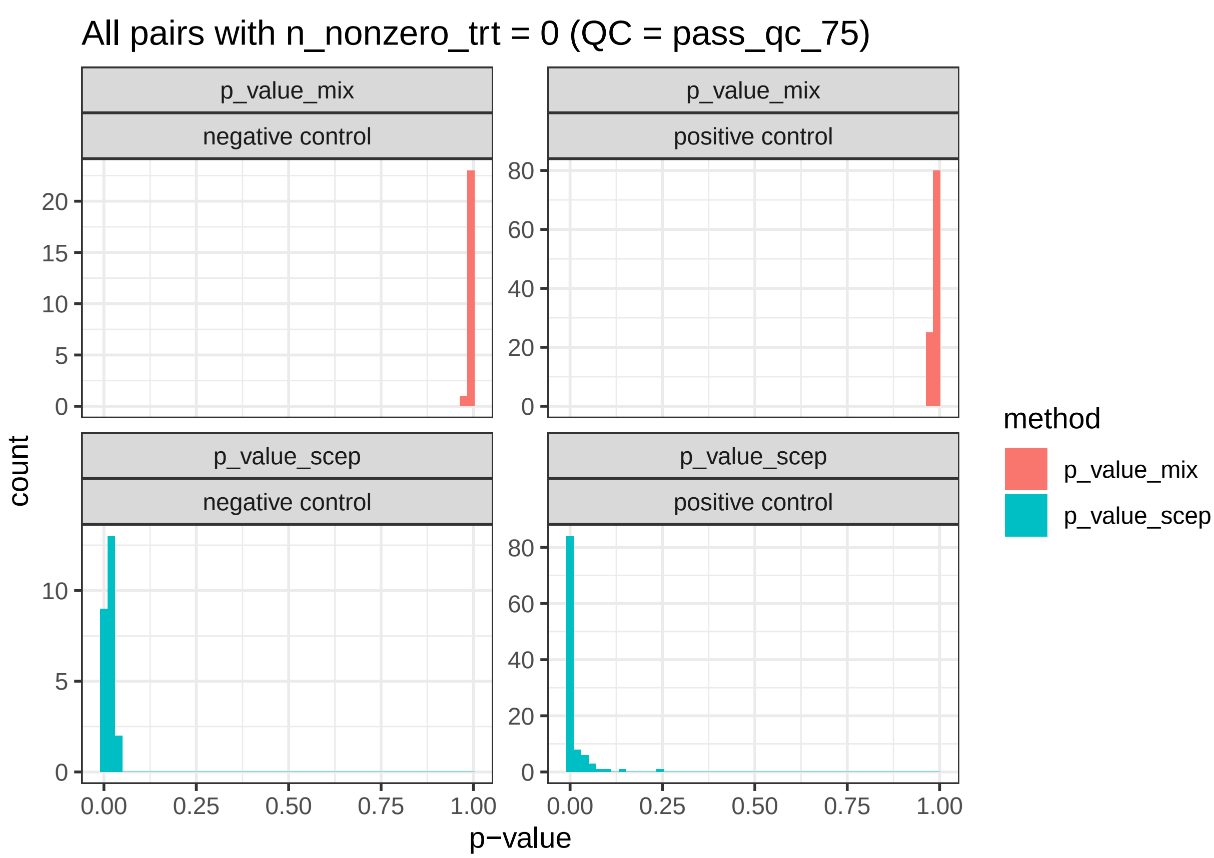


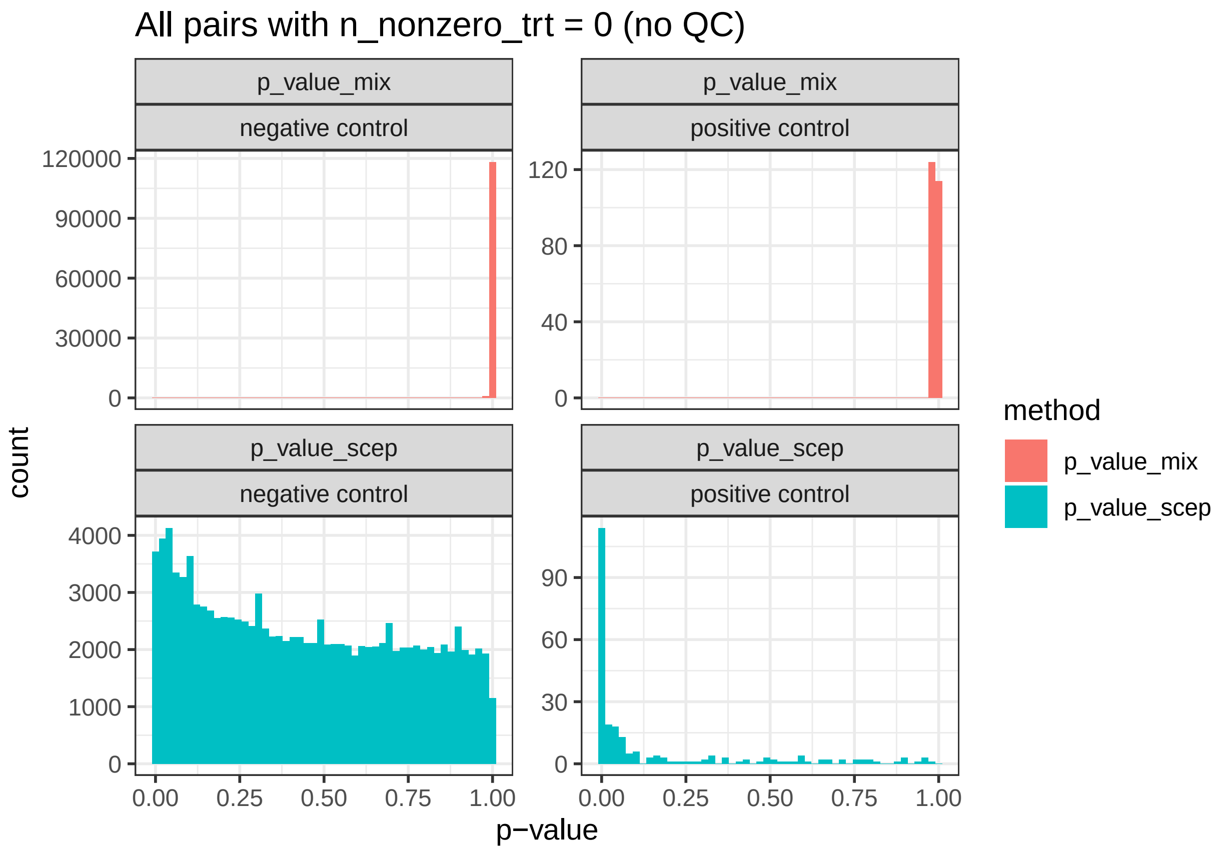
Figure 1: ROC analysis

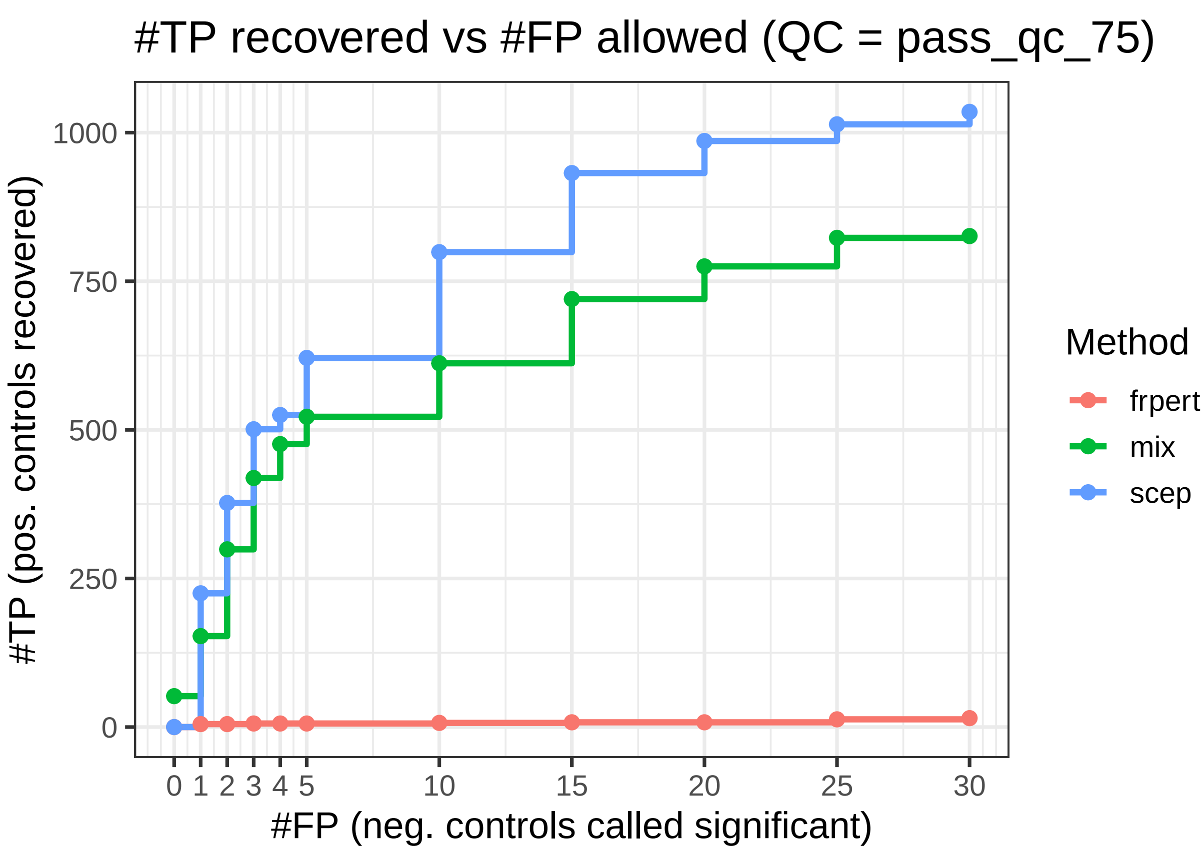


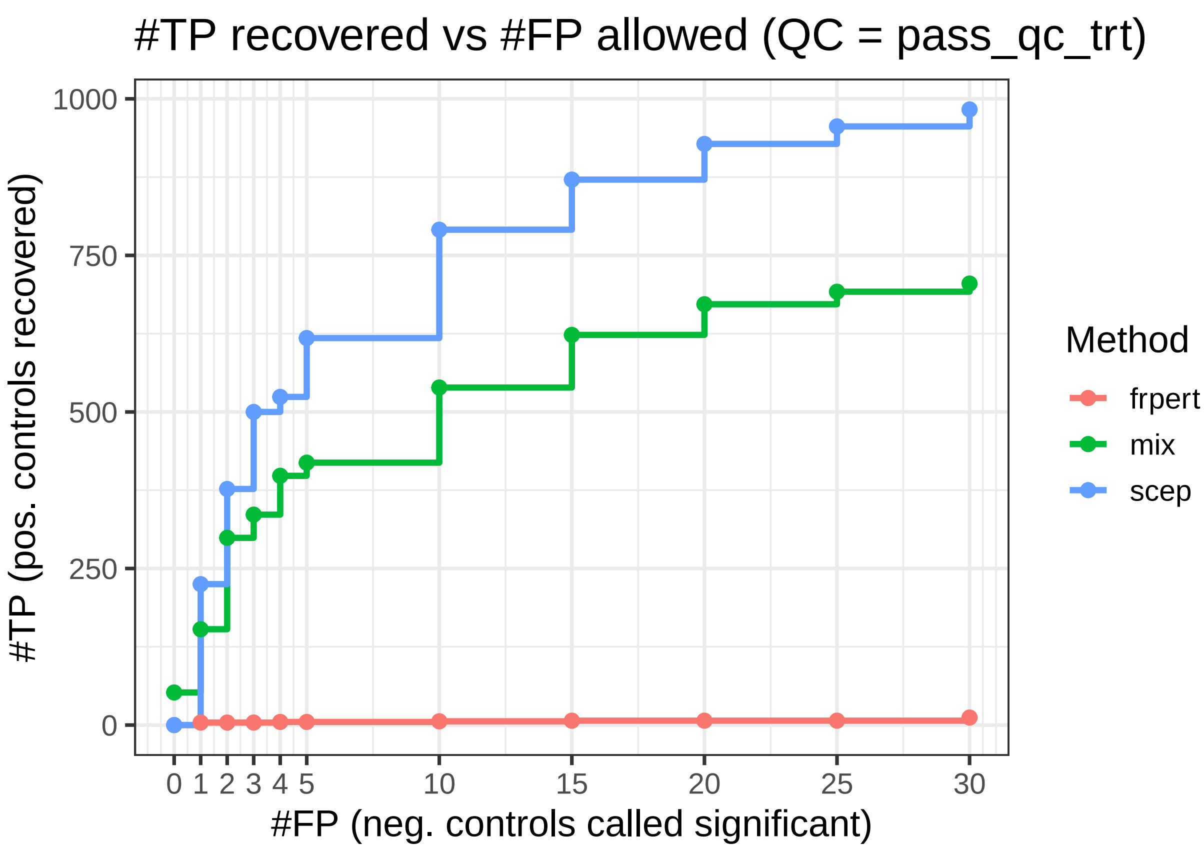
Separation of pos. and neg. (QC = pass_qc_def)
Recall
100%
75%
50%
25%
0%
0%
25%
50%
75%
100%
False Positive Rate
frpert (AUC = 0.716)
mix (AUC = 0.972)
scep (AUC = 0.967)

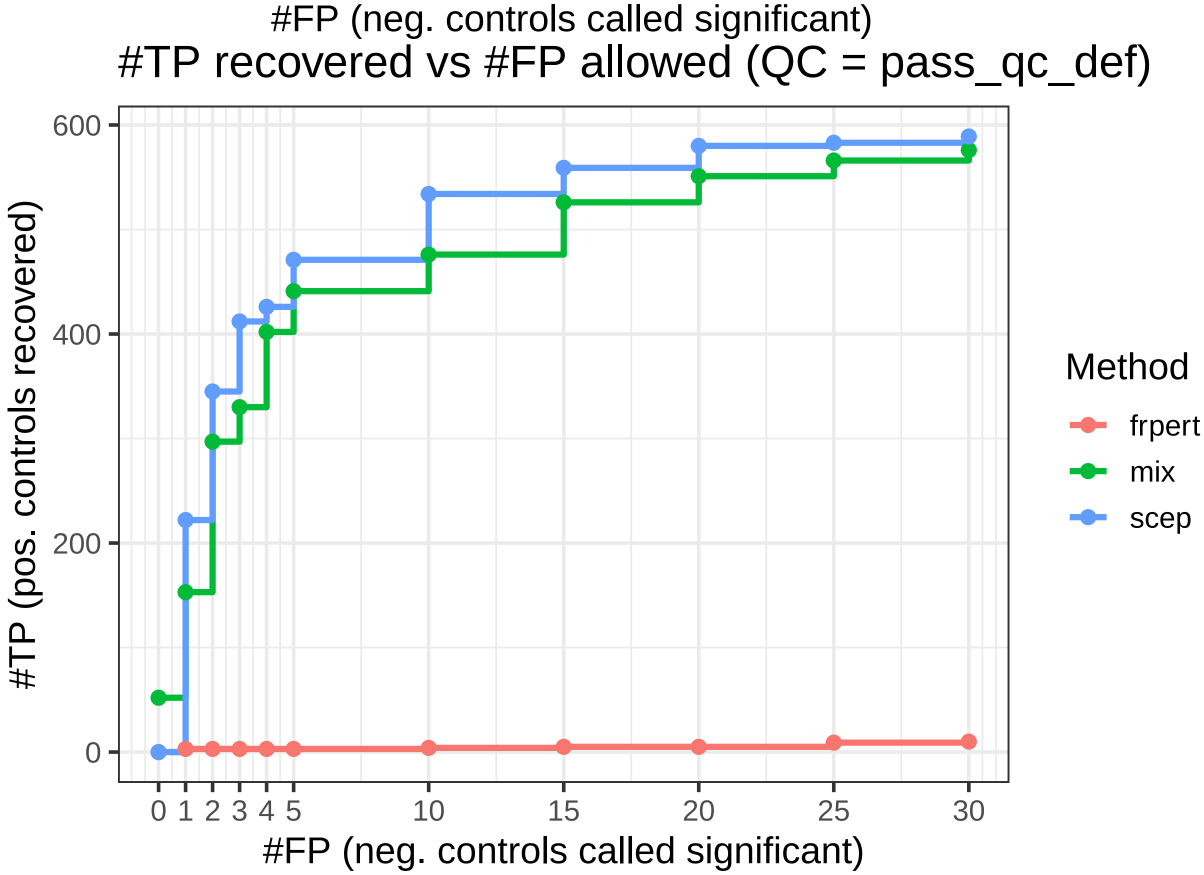


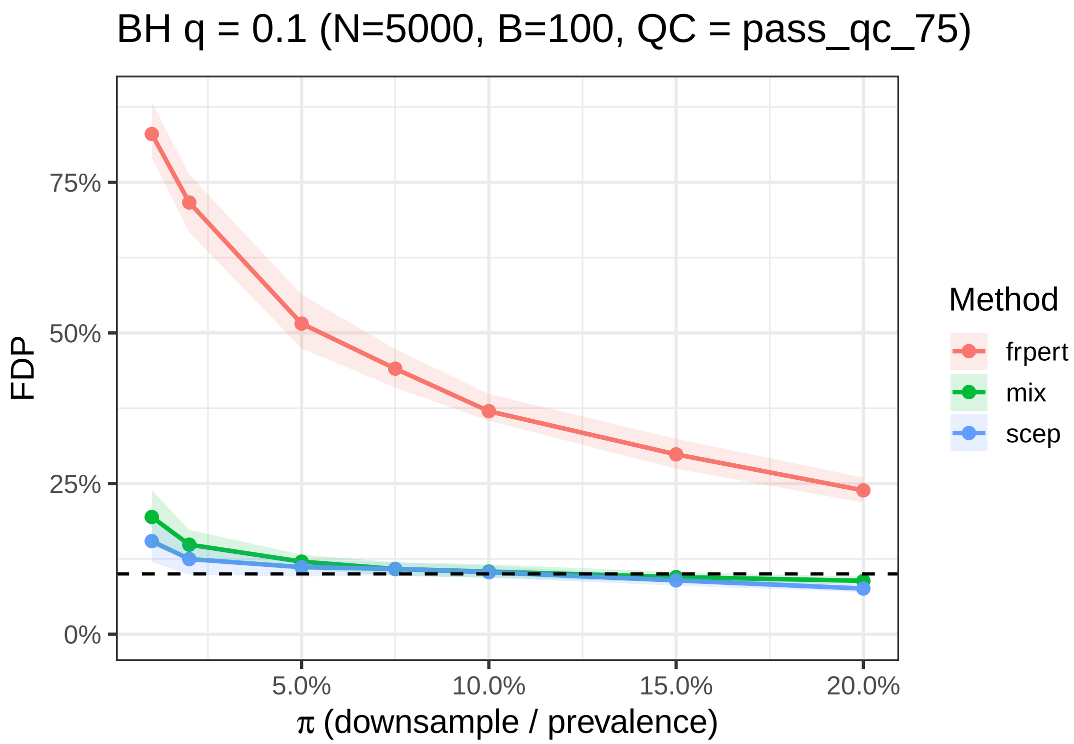
All pairs with n_nonzero_trt = 0 (QC = pass_qc_75)
p_value_mix
negative control
p_value_mix
positive control
p_value_scep
negative control
p_value_scep
positive control
count
p−value
0.00
0.25
0.50
0.75
1.00
method
p_value_mix
p_value_scep

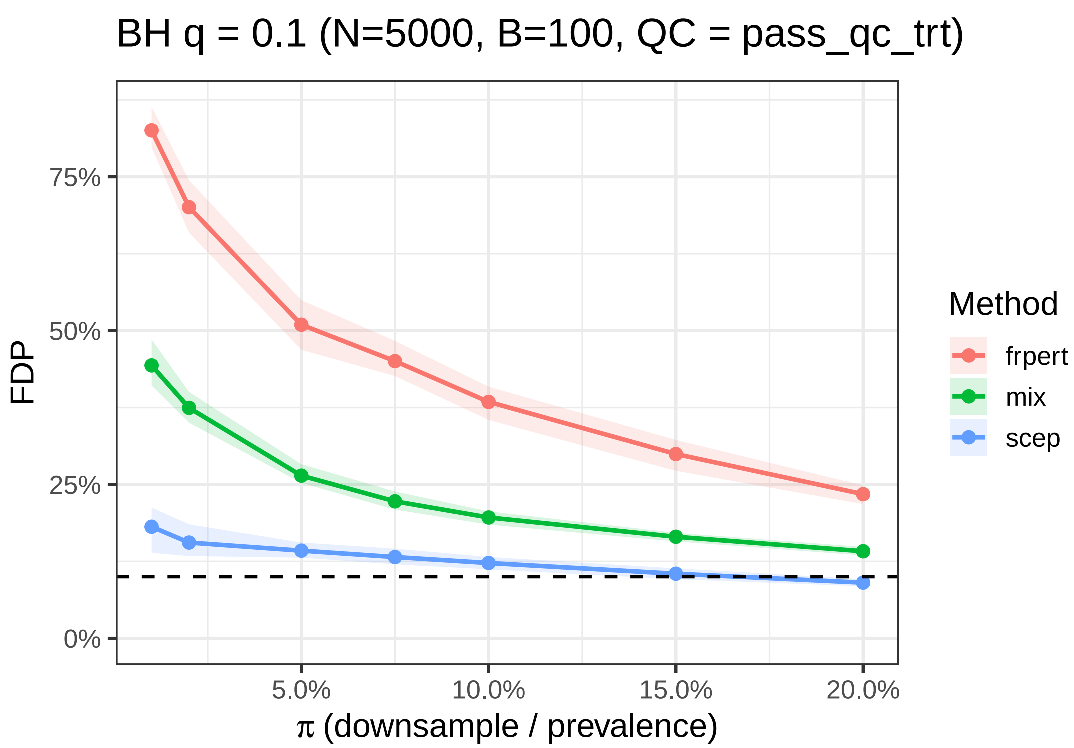

All pairs with n_nonzero_trt = 0 (no QC)
p_value_mix
negative control
p_value_mix
positive control
p_value_scep
negative control
p_value_scep
positive control
count
p−value
method
p_value_mix
p_value_scep
120000
90000
60000
30000
0
120
80
40
4000
3000
2000
1000
90
60
30
0.00
0.25
0.50
0.75
1.00

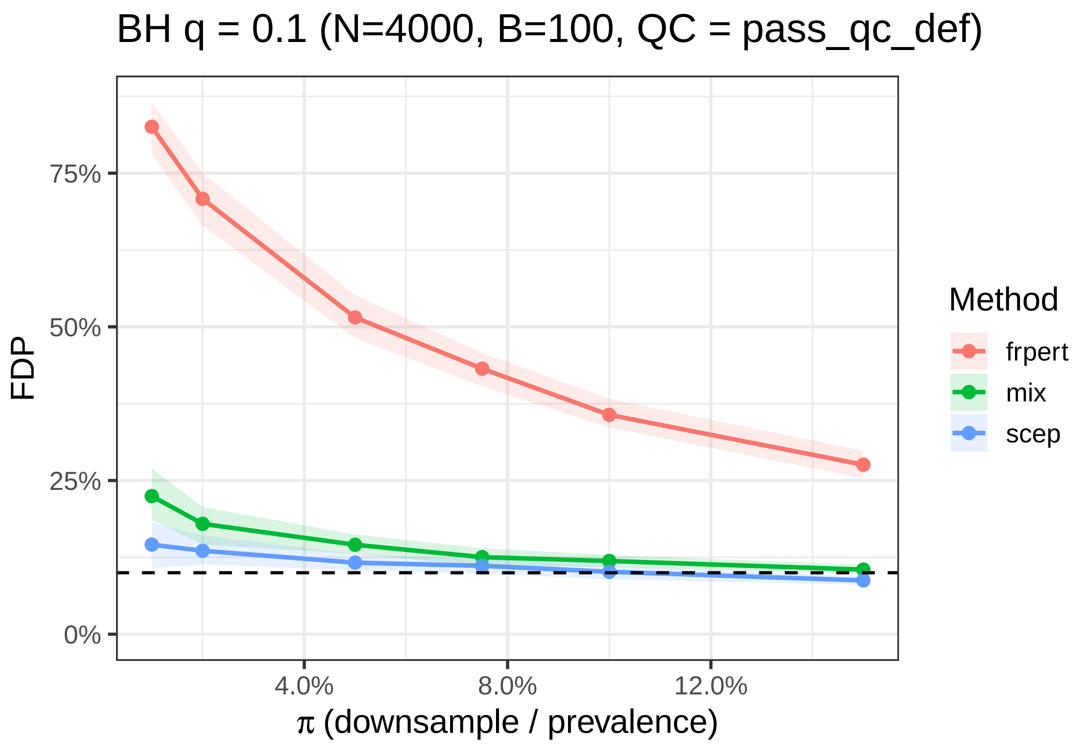

#TP recovered vs #FP allowed (QC = pass_qc_75)
#TP (pos. controls recovered)
#FP (neg. controls called significant)
Method
frpert
mix
scep
1000
750
500
250
0
0 1 2 3 4 5
10
15
20
25
30

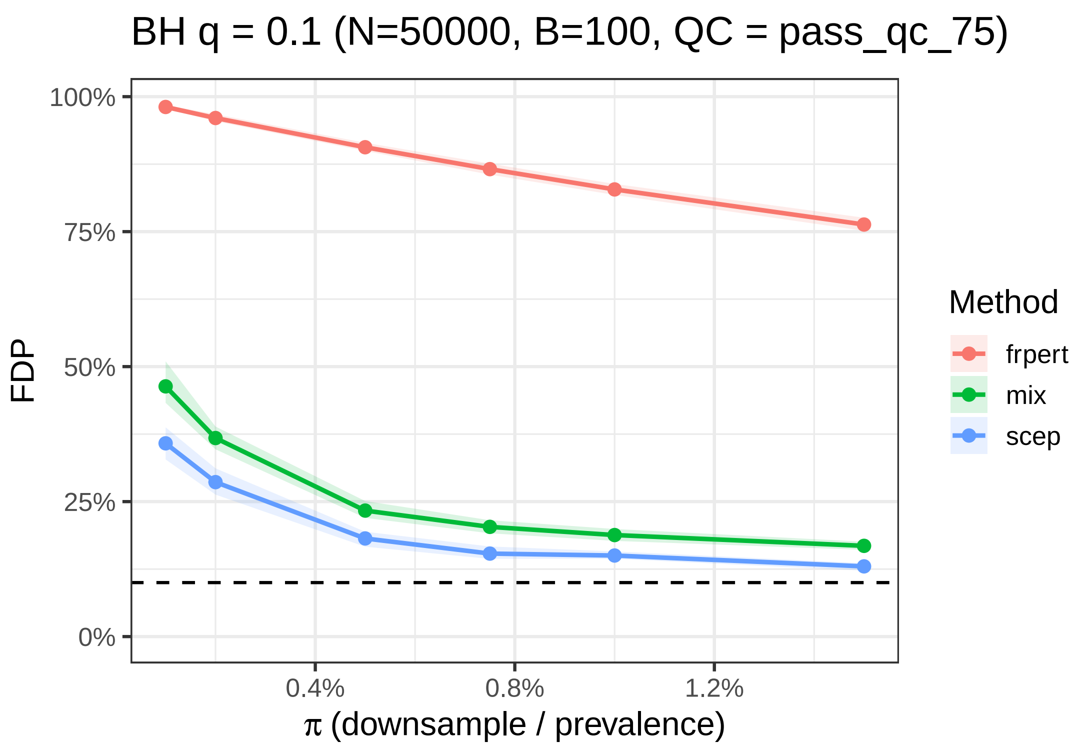

#TP recovered vs #FP allowed (QC = pass_qc_trt)
#TP (pos. controls recovered)
1000
750
500
250
0
0 1 2 3 4 5 10 15 20 25 30
#FP (neg. controls called significant)
Method
frpert
mix
scep

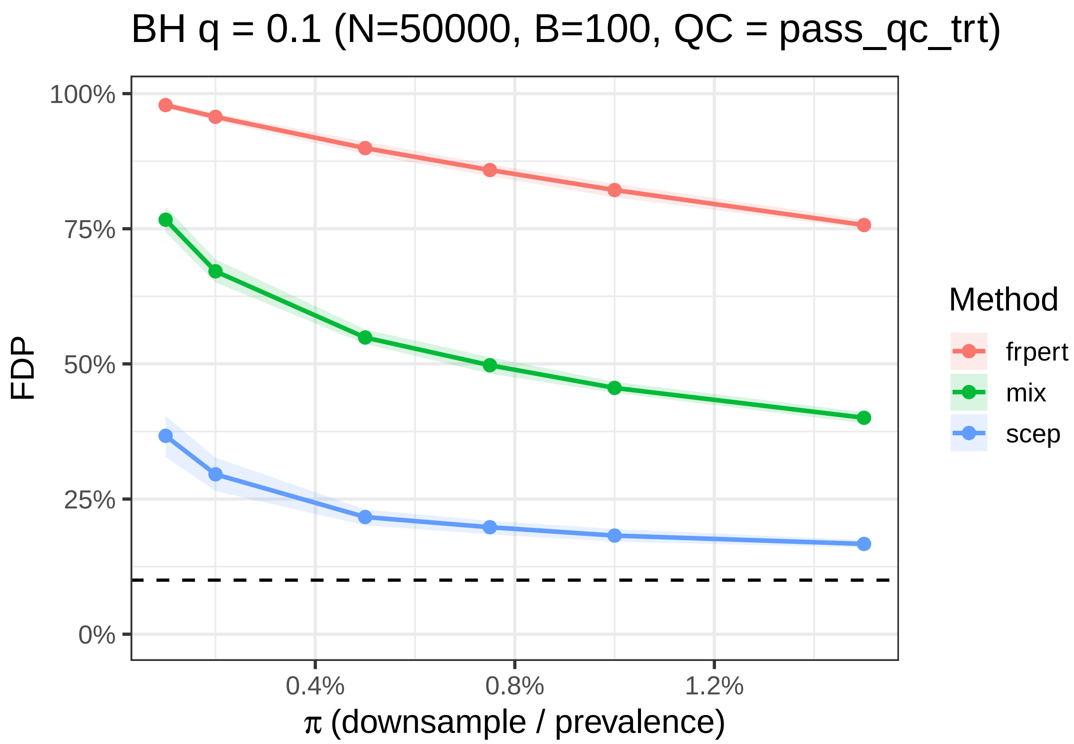

#TP recovered vs #FP allowed (QC = pass_qc_def)
#TP (pos. controls recovered)
600
400
200
0
0 1 2 3 4 5 10 15 20 25 30
#FP (neg. controls called significant)
Method
frpert
mix
scep

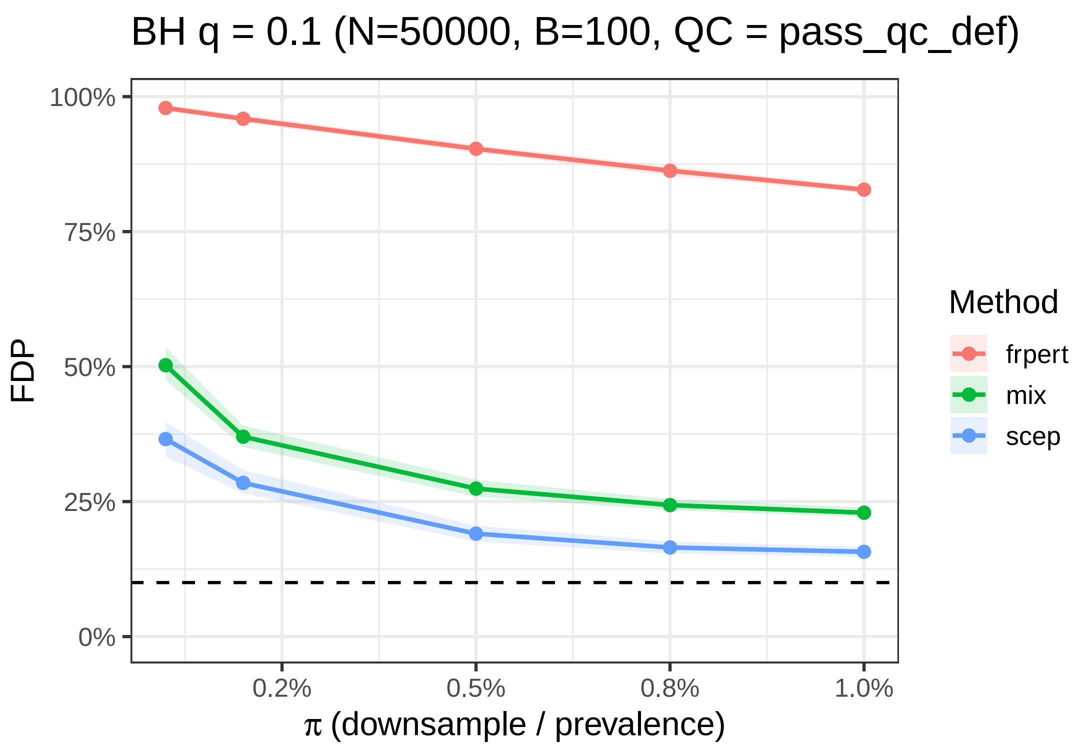

BH q = 0.1 (N=5000, B=100, QC = pass_qc_75)
FDP
75%
50%
25%
0%
5.0%
10.0%
15.0%
20.0%
π (downsample / prevalence)
Method
frpert
mix
scep

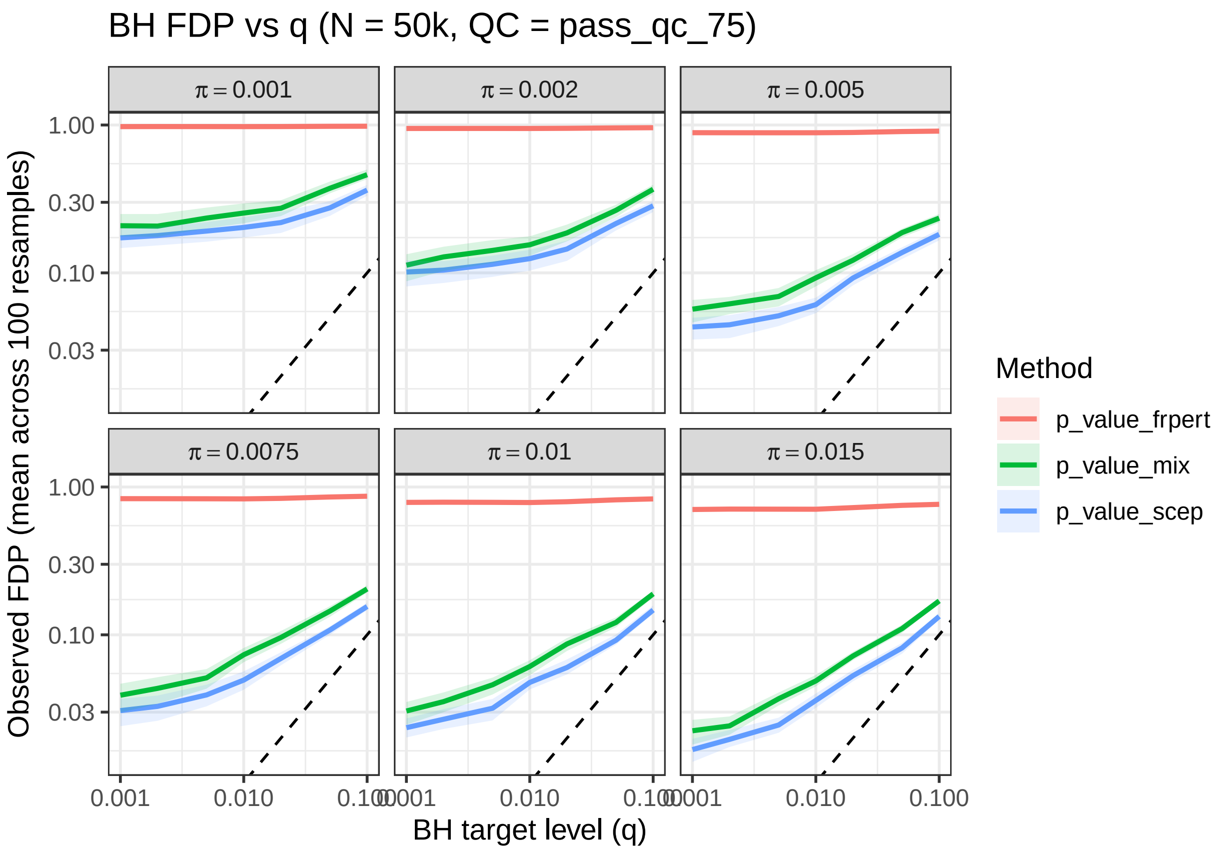

BH q = 0.1 (N=5000, B=100, QC = pass_qc_trt)
FDP
75%
50%
25%
0%
5.0%
10.0%
15.0%
20.0%
π (downsample / prevalence)
Method
frpert
mix
scep

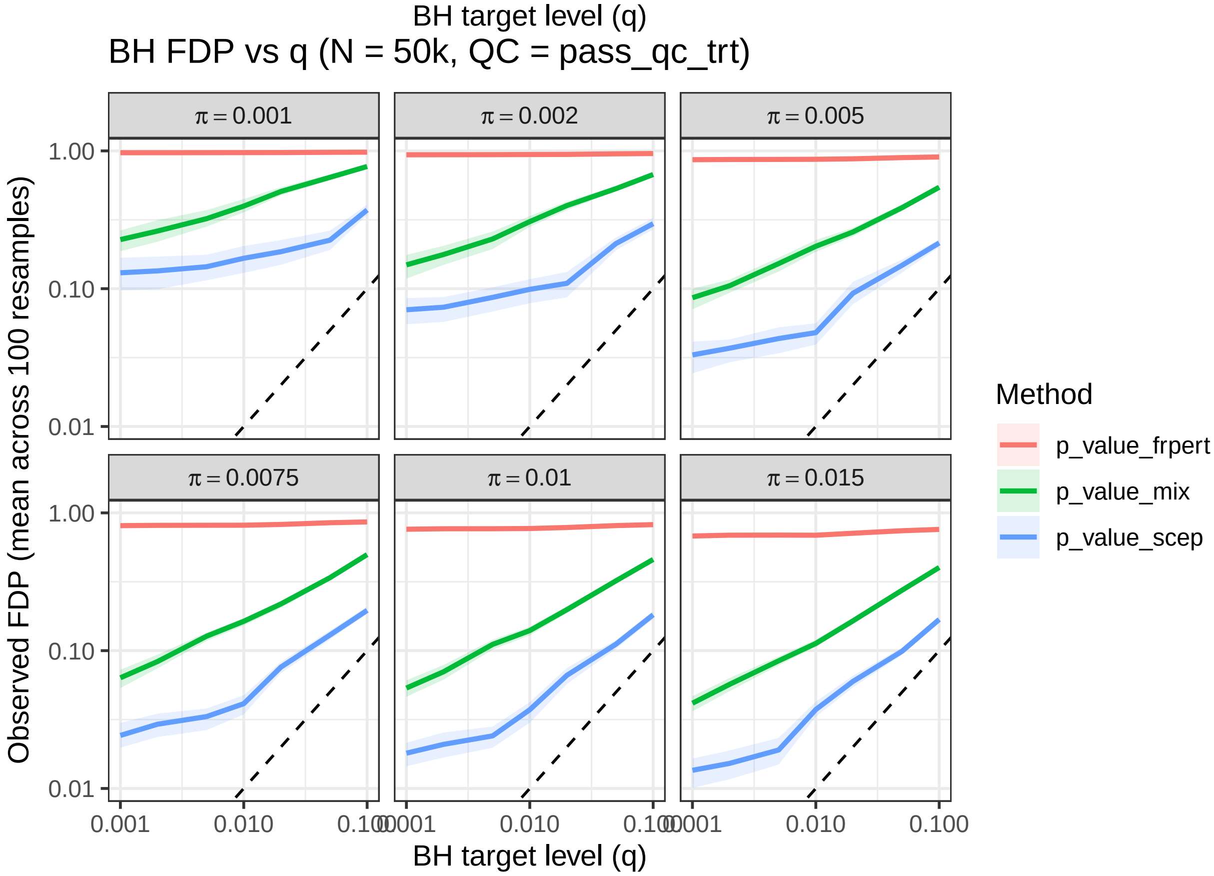

BH q = 0.1 (N=4000, B=100, QC = pass_qc_def)
FDP
75%
50%
25%
0%
4.0%
8.0%
12.0%
π (downsample / prevalence)
Method
frpert
mix
scep

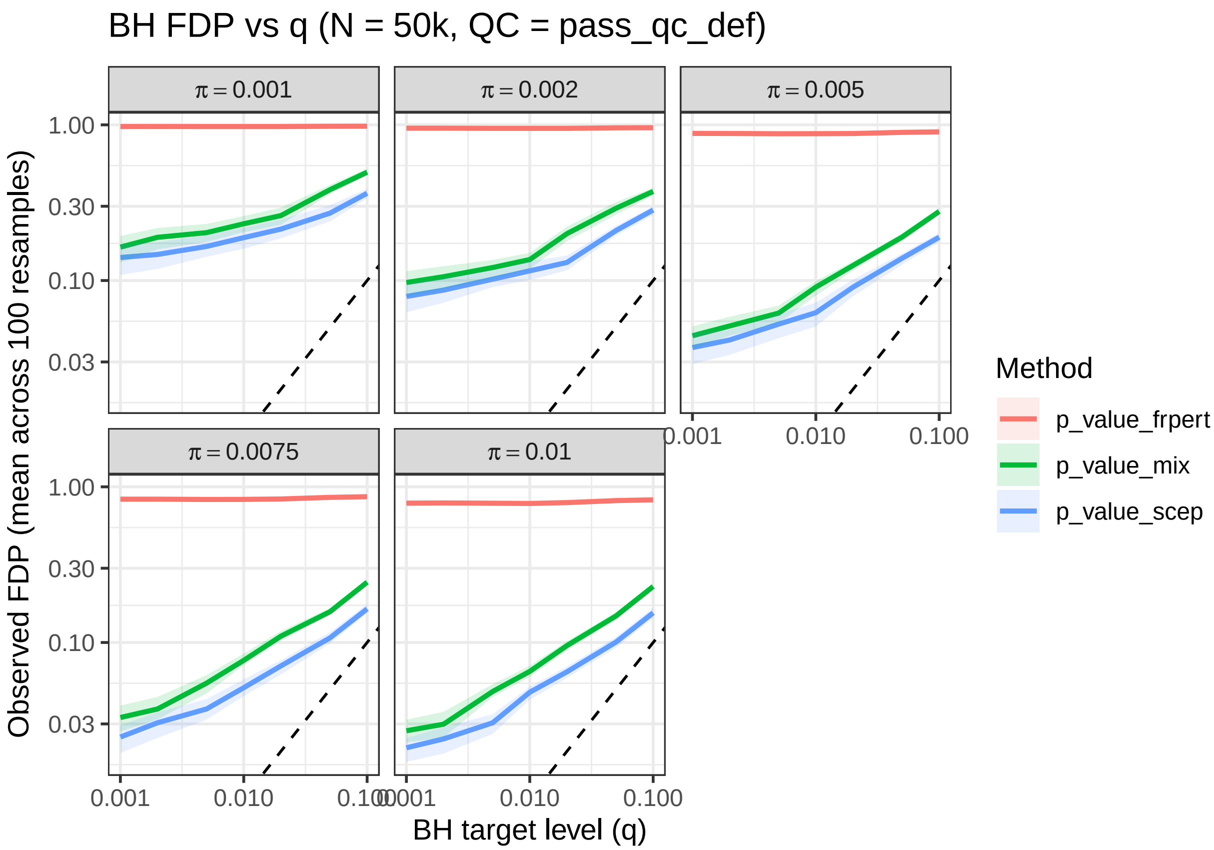

BH q = 0.1 (N=50000, B=100, QC = pass_qc_75)
FDP
100%
75%
50%
25%
0%
0.4%
0.8%
1.2%
π (downsample / prevalence)
Method
frpert
mix
scep


BH q = 0.1 (N=50000, B=100, QC = pass_qc_trt)
FDP
100%
75%
50%
25%
0%
0.4%
0.8%
1.2%
π (downsample / prevalence)
Method
frpert
mix
scep

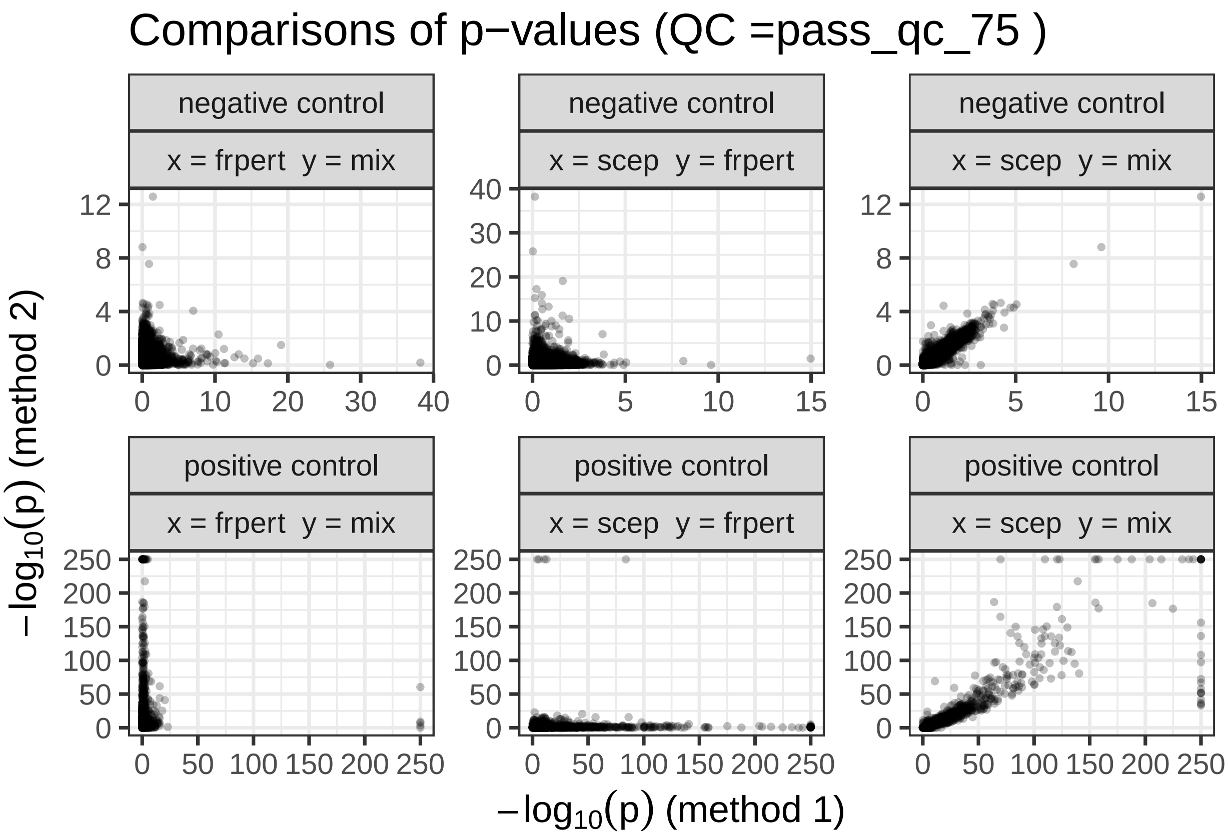

BH q = 0.1 (N=50000, B=100, QC = pass_qc_def)
FDP
100%
75%
50%
25%
0%
0.2%
0.5%
0.8%
1.0%
π (downsample / prevalence)
Method
frpert
mix
scep

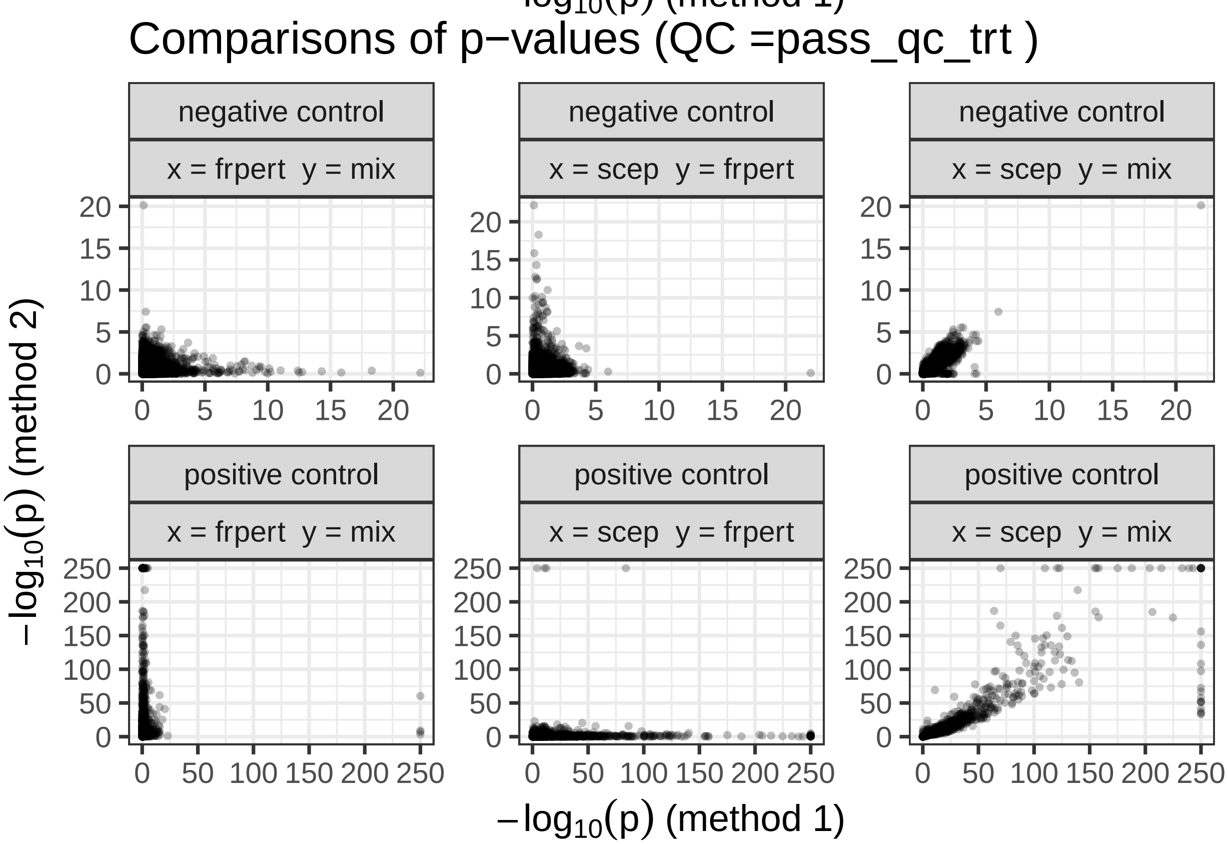

BH FDP vs q (N = 50k, QC = pass_qc_75)
π = 0.001
π = 0.002
π = 0.005
π = 0.0075
π = 0.01
π = 0.015
1.00
0.30
0.10
0.03
Observed FDP (mean across 100 resamples)
0.001
0.010
0.100
BH target level (q)
Method
p_value_frpert
p_value_mix
p_value_scep

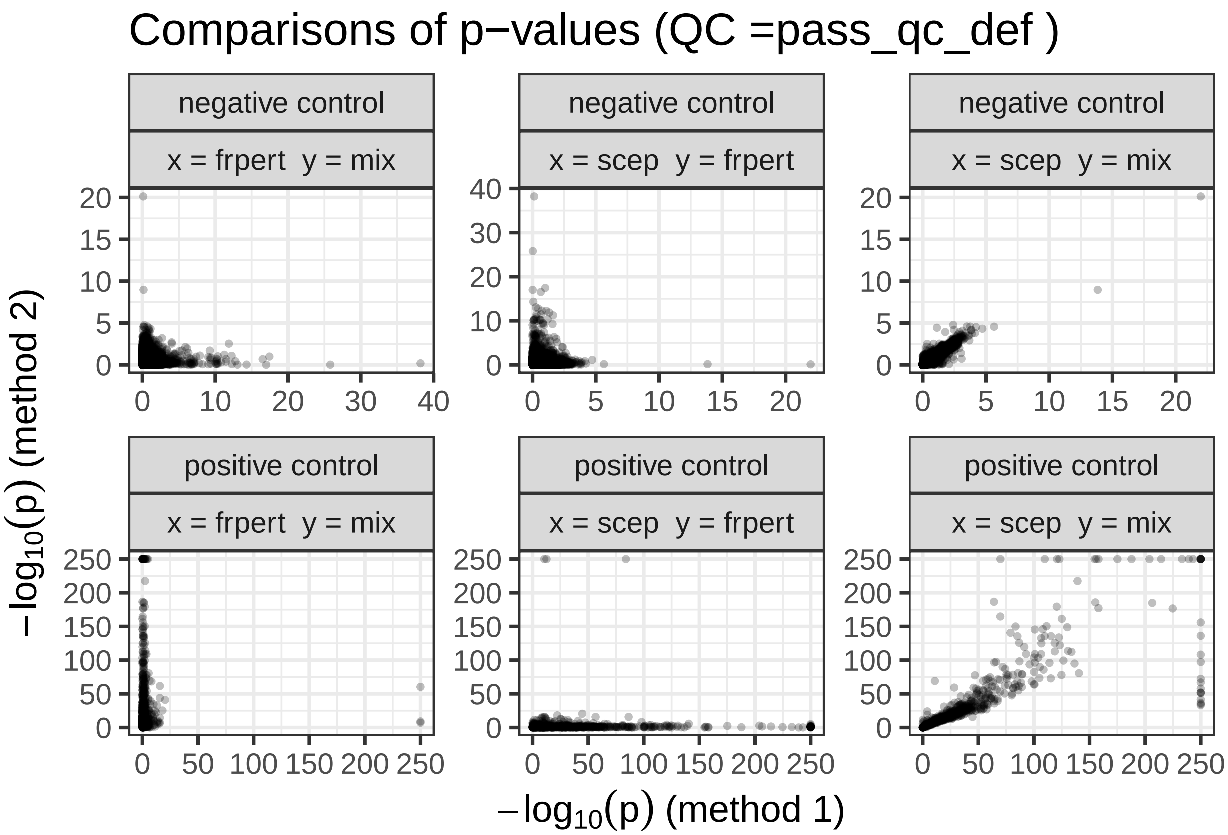

BH FDP vs q (N = 50k, QC = pass_qc_trt)
π = 0.001
π = 0.002
π = 0.005
π = 0.0075
π = 0.01
π = 0.015
1.00
0.10
0.01
Observed FDP (mean across 100 resamples)
0.001
0.010
0.100
BH target level (q)
Method
p_value_frpert
p_value_mix
p_value_scep



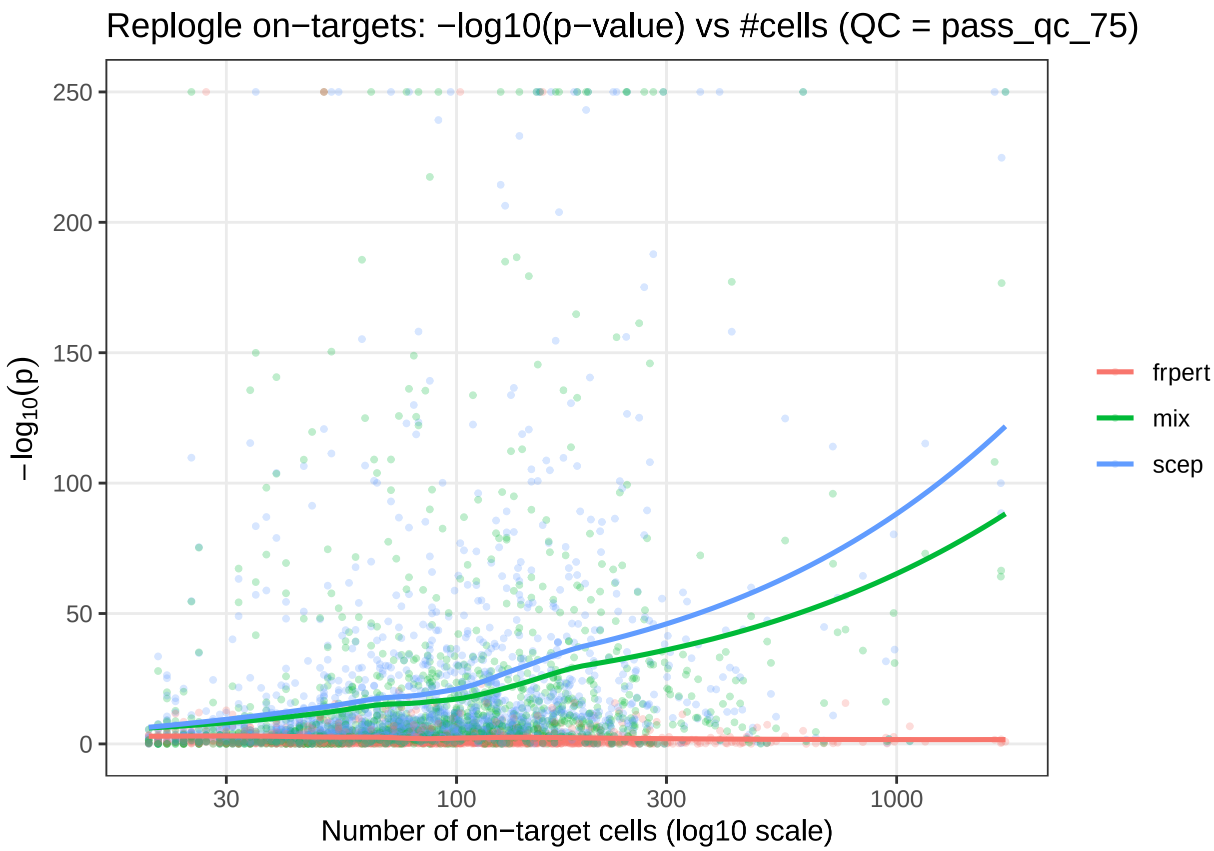

```
##          method1        method2         pair_type spearman          qc
## 1  p_value_scep     p_value_mix negative control     0.94  pass_qc_75
## 2  p_value_scep p_value_frpert negative control     0.01  pass_qc_75
## 3   p_value_mix p_value_frpert negative control     0.01  pass_qc_75
## 4  p_value_scep     p_value_mix positive control     0.93  pass_qc_75
## 5  p_value_scep p_value_frpert positive control    -0.01  pass_qc_75
## 6   p_value_mix p_value_frpert positive control     0.00  pass_qc_75
## 7  p_value_scep     p_value_mix negative control     0.93 pass_qc_trt
## 8  p_value_scep p_value_frpert negative control     0.01 pass_qc_trt
## 9   p_value_mix p_value_frpert negative control     0.01 pass_qc_trt
## 10 p_value_scep     p_value_mix positive control     0.96 pass_qc_trt
## 11 p_value_scep p_value_frpert positive control    -0.01 pass_qc_trt
## 12  p_value_mix p_value_frpert positive control    -0.01 pass_qc_trt
## 13 p_value_scep     p_value_mix negative control     0.94 pass_qc_def
## 14 p_value_scep p_value_frpert negative control     0.01 pass_qc_def
## 15  p_value_mix p_value_frpert negative control     0.01 pass_qc_def
## 16 p_value_scep     p_value_mix positive control     0.97 pass_qc_def
## 17 p_value_scep p_value_frpert positive control    -0.03 pass_qc_def
## 18  p_value_mix p_value_frpert positive control    -0.03 pass_qc_def
```


Comparisons of p−values (QC =pass_qc_75)
negative control
x = frpert y = mix
negative control
x = scep y = frpert
negative control
x = scep y = mix
positive control
x = frpert y = mix
positive control
x = scep y = frpert
positive control
x = scep y = mix
$-\log_{10}(p)$ (method 2)
$-\log_{10}(p)$ (method 1)

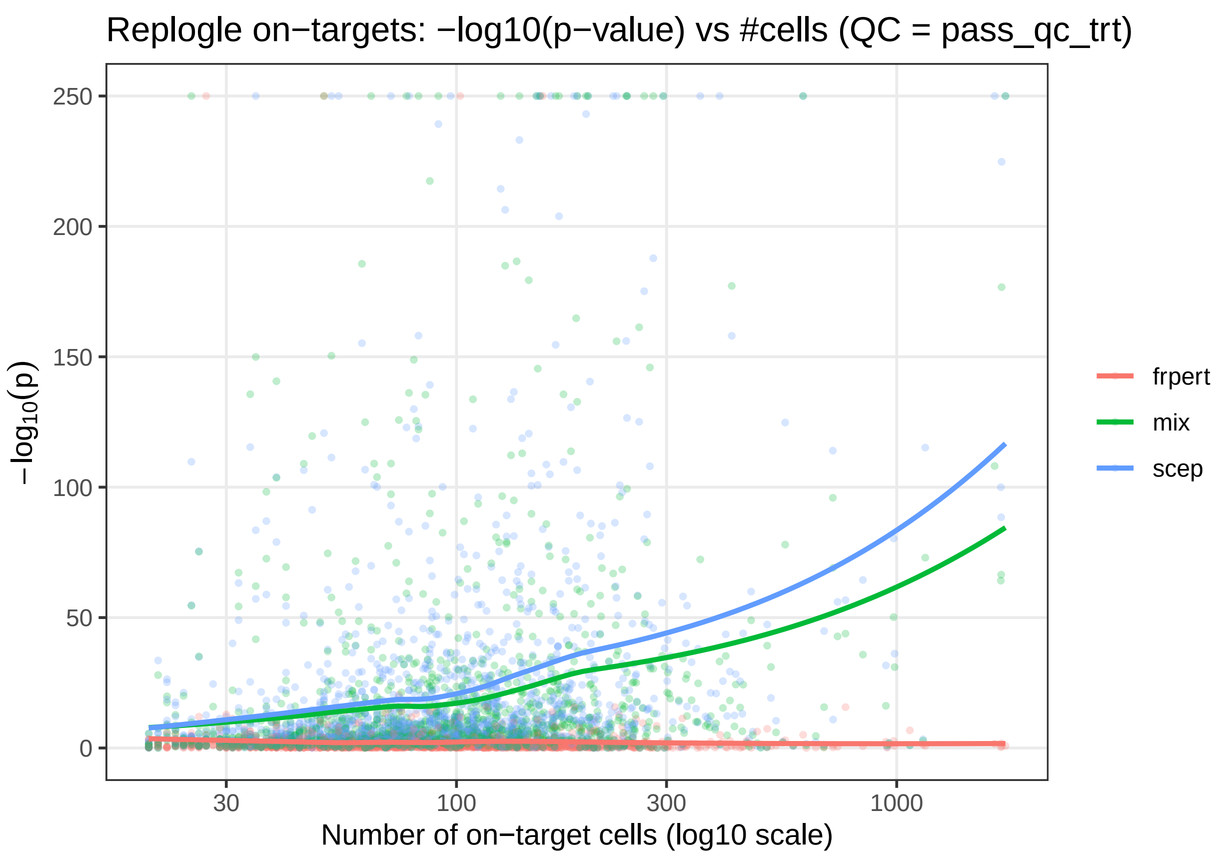

Comparisons of p−values (QC =pass_qc_trt)
negative control
x = frpert y = mix
negative control
x = scep y = frpert
negative control
x = scep y = mix
positive control
x = frpert y = mix
positive control
x = scep y = frpert
positive control
x = scep y = mix
$-\log_{10}(p)$ (method 2)
$-\log_{10}(p)$ (method 1)

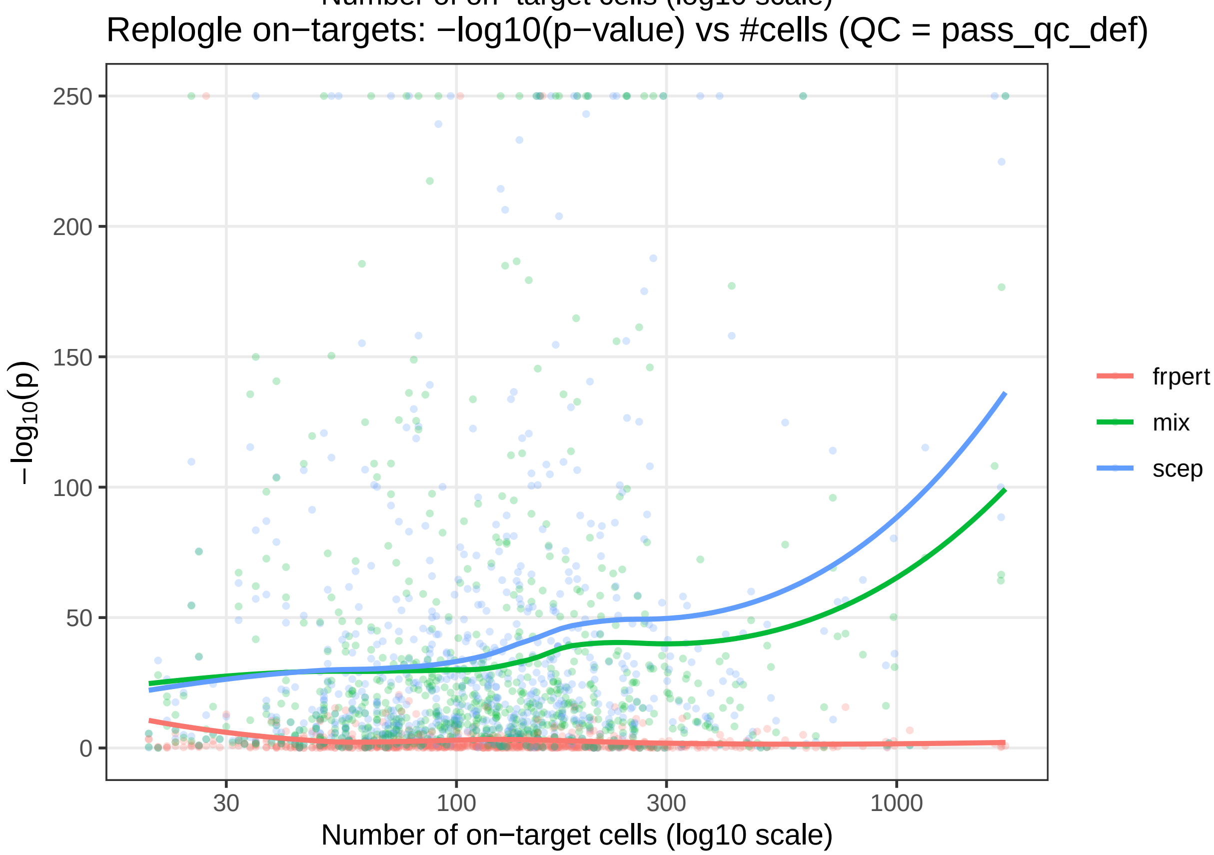


Comparisons of p−values (QC =pass_qc_def)
negative control
x = frpert y = mix
negative control
x = scep y = frpert
negative control
x = scep y = mix
positive control
x = frpert y = mix
positive control
x = scep y = frpert
positive control
x = scep y = mix
−log10(p) (method 2)
−log10(p) (method 1)



Replogle on−targets: −log10(p−value) vs #cells (QC = pass_qc_75)
−log10(p)
Number of on−target cells (log10 scale)
frpert
mix
scep

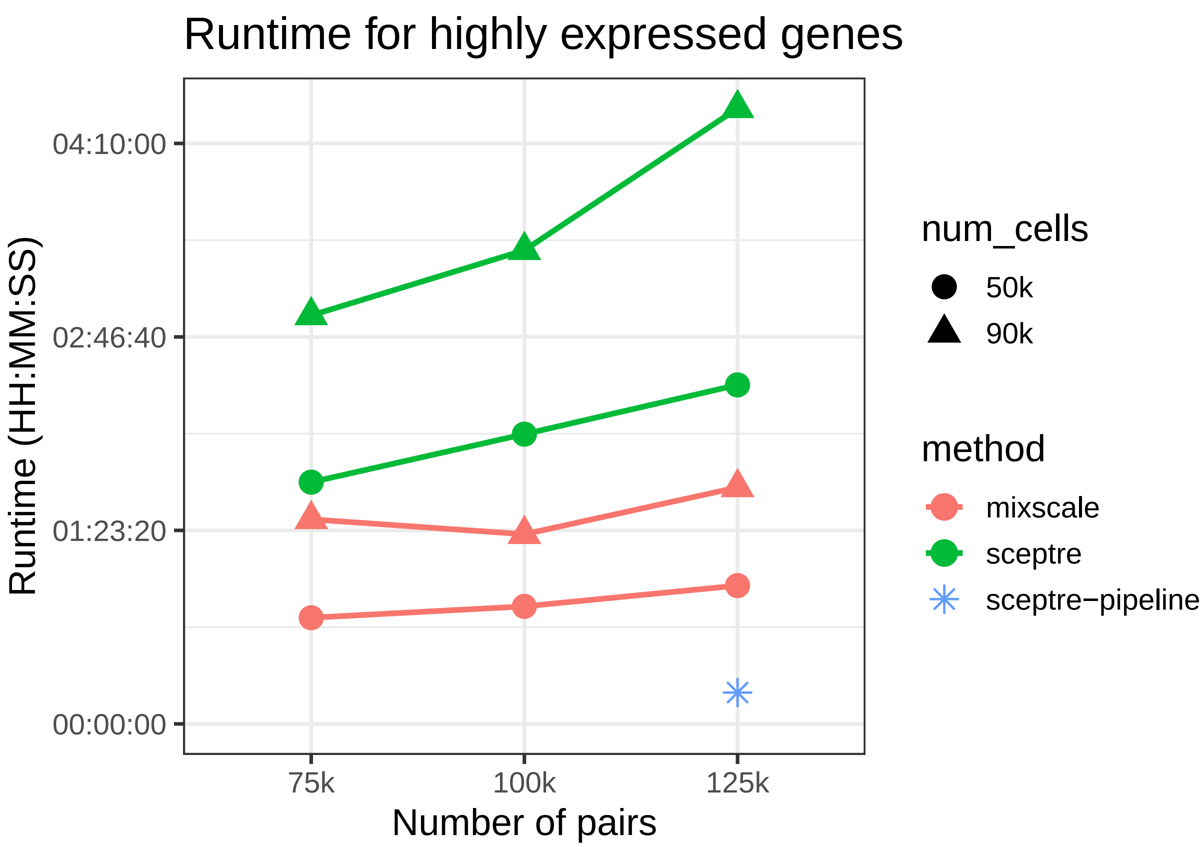

Replogle on−targets: −log10(p−value) vs #cells (QC = pass_qc_trt)
250
200
150
100
50
0
−log10(p)
30
100
300
1000
Number of on−target cells (log10 scale)
frpert
mix
scep


Replogle on−targets: −log10(p−value) vs #cells (QC = pass_qc_def)
250
200
150
100
50
0
−log10(p)
30
100
300
1000
Number of on−target cells (log10 scale)
frpert
mix
scep

```r
# compute
# make plot of times, add in pipeline
# and repeat for memory

run_name = "run_repl_mix_scep_highly_expressed_seq"
scep_pipe_run = "replogle-rd7_comp_ngenes=560_ntargets=225_ncells=90k_n_nonzero_p=0.75"
base_path_comp = file.path(.get_config_path("LOCAL_BENCHMARKING_DIR"),
                                  "association/computational")
scep_pipe = read_tsv(
  file.path(base_path_comp, "outputs/sceptre-pipeline", scep_pipe_run, "tracing/trace.tsv"),
  show_col_types = FALSE
)


## hard coded for now!!
# assoc_start_process = "prepare_association_analysis"
# start_time = dplyr::filter(scep_pipe, process == assoc_start_process) |>
#   pull(start)
# end_time = max(scep_pipe$complete)
# elapsed = end_time - start_time
#
# # max memory per node all under 1GB
# scep_pipe |>
#   dplyr::filter(start >= start_time) |>
#   pull(peak_rss) |>
#   sort()

pair_data = data.frame(
  ngenes = c(430, 500, 560),
  ntargets = c(175, 200, 225),
  npairs = c(75, 100, 125)
)

trace_file = read_tsv(
  file.path(base_path_comp, "outputs", run_name, "trace.txt"),
  show_col_types = FALSE
) |>
  tidyr::extract(
    tag,
    into = c("ngenes", "ntargets"),
    regex = "ngenes=([0-9]+)_ntargets=([0-9]+)",
    remove = FALSE,
    convert = TRUE
  ) |>
  transmute(
    method = gsub("_computational", "", process, ignore.case = TRUE) |> tolower(),
    peak_rss, realtime,
    num_cells =  sub(".*ncells=([^_]+).*", "\\1", tag), ngenes, ntargets
  ) |>
  mutate(
    peak_rss_gb = case_when(
      str_detect(peak_rss, regex("\\bGB\\b", ignore_case = TRUE)) ~ parse_number(peak_rss),
      str_detect(peak_rss, regex("\\bMB\\b", ignore_case = TRUE)) ~ parse_number(peak_rss) / 1024,
```

```
      TRUE ~ NA_real_
    )
  ) |>
  mutate(
    h = parse_number(str_extract(realtime, "\\d+(?=h)")),
    m = parse_number(str_extract(realtime, "\\d+(?=m)")),
    s = parse_number(str_extract(realtime, "\\d+(?=s)")),
    h = coalesce(h, 0),
    m = coalesce(m, 0),
    s = coalesce(s, 0),
    realtime_sec = 3600*h + 60*m + s,
    realtime_min = realtime_sec / 60
  ) %>%
  select(-h, -m, -s) |>
  left_join(pair_data, by = c("ngenes", "ntargets"))


# helper: seconds -> "HH:MM:SS"
sec_to_hms <- function(x) {
  x <- round(x)
  h <- x %/% 3600
  m <- (x %% 3600) %/% 60
  s <- x %% 60
  sprintf("%02d:%02d:%02d", h, m, s)
}

# df_plot should have: realtime_sec, method, num_cells, npairs
x_levels <- c(75, 100, 125)
x_labels <- paste0(x_levels, "k")

df_plot2 <- trace_file %>%
  mutate(
    npairs_cat = factor(npairs, levels = x_levels, labels = x_labels),
    num_cells  = factor(num_cells)
  )

# custom point: (125k, 00:13:30) = 810 sec
df_pipeline <- tibble(
  npairs_cat   = factor("125k", levels = x_labels),
  realtime_sec = 13 * 60 + 30,
  method       = "sceptre-pipeline"
)

ggplot(df_plot2, aes(
  x = npairs_cat, y = realtime_sec,
  color = method, shape = num_cells,
  group = interaction(method, num_cells)
)) +
  geom_line(linewidth = 1) +
  geom_point(size = 4) +
  geom_point(
    data = df_pipeline,
    aes(x = npairs_cat, y = realtime_sec, color = method),
```

```
    inherit.aes = FALSE,
    shape = 8, size = 3
  ) +
  scale_y_continuous(
    limits = c(0, NA),                  # include 00:00:00
    labels = sec_to_hms
  ) +
  labs(
    x = "Number of pairs",
    y = "Runtime (HH:MM:SS)",
    color = "method",
    shape = "num_cells",
    title = "Runtime for highly expressed genes"
  ) +
  theme_bw(base_size = 14)
```




```
ggplot(df_plot2, aes(
  x = npairs_cat, y = peak_rss_gb,
  color = method, shape = num_cells,
  group = interaction(method, num_cells)
)) +
  geom_line(linewidth = 1) +
  geom_point(size = 4) +
  scale_y_log10() +
  labs(
    x = "Number of pairs",
    y = "Peak RSS (log10 scale)",
```

```r
    color = "method",
    shape = "num_cells",
    title = "Memory for highly expressed genes"
  ) +
  theme_bw(base_size = 14)
```